

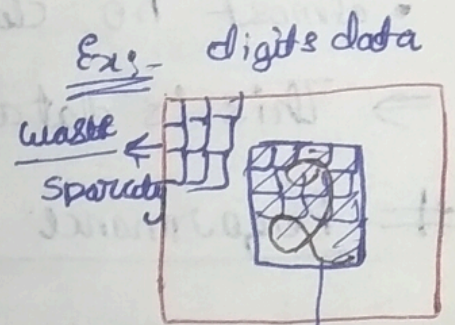
Principal Component Analysis (PCA)

Curse of Dimensionality :-

The curse of dimensionality refers to the problems that arise when the number of features (dimensions) become very large.

→ as dimension increases :

- data becomes sparse
- distance measures become meaningless
- model performs worse, not better.



these features are enough to predict

Simple intuition

1. 1D :- A line with few data points can cover it well.
Ex: - Purse lost in the road

2D :- A square/plane need more data to fill.
Ex: - Purse lost in the big building playground

3D :- A huge space, Data points are far apart
Ex: - same purse lost in the huge building.

Real world Example: - (few data, huge space) $\Rightarrow$ Sparsity analogy

i) Image data :-

- $224 \times 224 \times 3 \Rightarrow 150,528$ features
- distance based model struggles

ii) Text data
Bag of words :-

- 10,000 + features
- Sparse vectors.

Why is this a curse?

Because many ML algorithms assume:

- nearby points are similar
- distance is meaningful

In higher dimensions: Nearest Neighbor & farthest neighbor

Key Problem:- Data Sparsity

Data sparsity means that the data points are very far apart from each other in a high dimensional space.

→ As the number of features increases, the space grows exponentially but the no. of data points remains the same.

→ Result: - most of the space is empty

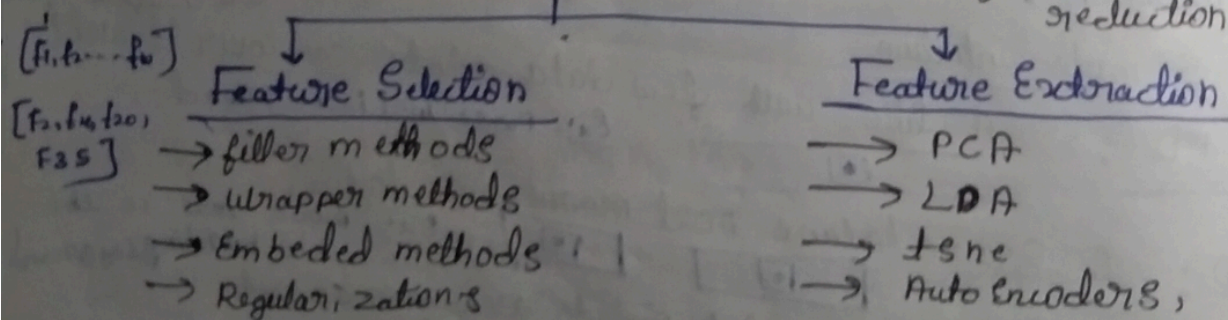
→ High Dimension (10, 100, 1000)

- Space becomes huge
- points are isolated
- almost no close neighbors

⇒ This is data sparsity

Performance decreases, # Expensive Computation

Solutions to Curse of Dimensionality:- i) Dimensionality reduction



What is PCA [Principal Component Analysis]

PCA [Principal Component Analysis] is an unsupervised dimensionality reduction technique used to transform a high-dimensional dataset into a lower-dimensional space while preserving maximum variance (information).